Automatic Retail Product Checkout using Detection and Embedding-Based Retrieval

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***Abstract*—This project presents an automated retail checkout system for identifying multiple products in cluttered shelf images. We explore and compare three approaches: a baseline YOLOv8 model for direct detection and classification, a hybrid YOLOv8 + VAE pipeline for embedding-based retrieval, and a YOLOv8 + DINOv2 pipeline for improved feature representation. In all approaches, YOLOv8 is used to detect and crop individual products from input images. For the hybrid methods, cropped images are converted into embeddings and matched against a database using k-nearest neighbor (k=5) similarity search with majority voting. While the VAE-based approach provides a simple latent representation, it struggles to distinguish visually similar products. In contrast, the DINOv2-based approach produces more discriminative embeddings, resulting in better retrieval performance. The comparison demonstrates that embedding-based retrieval improves flexibility over direct classification, and the YOLOv8 + DINOv2 pipeline achieves the best overall performance while allowing seamless addition of new products without retraining.**

***Index Terms*—YOLOv8, DINOv2, VAE, Vision Transformer, Vector Embeddings, Similarity Search, k-NN, Retail Checkout System**

I. Introduction

Making retail checkout automatic has turned into a big research topic lately because stores really need faster and more accurate ways to handle customers. But identifying products in a real store is a mess. You have crowded shelves, items hidden behind each other, and dozens of different products that look almost exactly the same. These problems make it tough for standard computer vision systems to work reliably, especially when a store is constantly adding new items to the shelves.

We have seen a lot of progress in deep learning for retail, but it is still not perfect. Early on, everyone used barcodes or RFID, but those are slow and require a person to physically move every item, which just does not scale well [1]. When deep learning came along, models like YOLO and Faster R-CNN became the go-to tools because they can pick out visual features straight from an image [6]–[8]. They work great in a lab, but in a real store, they often struggle when products are piled on top of each other or look too similar [5]. Also, there is a big gap between training and reality; a model might know what a soda bottle looks like in a clean photo, but it gets confused at a messy checkout counter [2].

To try and fix this, people have tried everything from creating fake training data to using attention models that help the computer focus better [3], [9]. There are even huge datasets like RPC now to help simulate these scenes [10]. The problem is that most of these still rely on basic classification. That means if the store gets a new product, you have to retrain the whole model, which is a huge pain. Because of that, research is moving toward embeddings. Instead of just naming an object, the model creates a visual fingerprint (an embedding) so it can just search for a match in a database [11]–[13]. This is much more flexible for a store that changes its stock all the time.

In this project, we wanted to see if we could build a system that handles these similar items and messy scenes without needing constant retraining. Our goal was to make something that actually works in a changing retail environment.

Our solution uses a mix of object detection and a similarity search. We use YOLOv8 to find the products and crop them out. Then, we turn those crops into vector embeddings. We started with a Variational Autoencoder (VAE), but it just was not detailed enough to tell the difference between similar-looking boxes. So, we switched to DINOv2. Since it uses a Vision Transformer with self-attention, it is much better at picking up the small details and context of an image. To actually name the product, we use a k-nearest neighbors (k-NN) search and a majority vote to find the best match.

The rest of this report is laid out to show how we built and tested the system. We go over what other people have done, explain our specific methodology, show the data we used, and then break down the results to see which approach actually worked best.

II. Related Work

Occluded Retail Product Detection. Chan et al. [3] introduce an attention-reweighted Faster R-CNN model to handle occlusions in retail environments. The model uses region proposals followed by feature reweighting through attention mechanisms, allowing it to better handle overlapping products. Experiments on the Snacks-MIT dataset show that the approach achieves an mAP of 84.47. However, the method increases model complexity and still depends heavily on labeled data, making it less flexible for large-scale retail deployment.

Enhanced Self-Checkout System for Retail Based on Improved YOLOv10 Recent work using improved YOLOv10 [5] enhances detection accuracy and speed for retail checkout systems. This work proposes an improved retail checkout system using YOLOv10 to handle challenges like clutter, occlusion, and visually similar products. The authors enhance the model by combining ideas from YOLOv8 and YOLOv10 and adding attention mechanisms such as EMA along with lightweight convolution modules. The system is trained on the RPC dataset and shows better detection performance in complex retail scenes, although it requires careful tuning and computational resources.

RPC Dataset for Retail Checkout. Wei et al. [10] introduce the RPC dataset, a large-scale dataset for automatic checkout containing 200 product categories and over 83,000 images. It includes both single-product and real-world checkout images with clutter and occlusion. The authors also propose a cross-domain approach using synthetic image generation and Cycle-GAN for domain adaptation. Despite strong performance, challenges remain in handling dense scenes and visually similar products.

DINO for Visual Representation Learning. Caron et al. [12] propose DINO, a self-supervised method for Vision Transformers that learns visual features without labeled data using a teacher–student framework. It captures meaningful semantic information and performs well even with simple classifiers. However, it requires careful tuning and can be computationally expensive.

YOLO + SimCLR Hybrid Approaches. James et al. [14] propose a two-stage system combining YOLOv8-based detection with SimCLR self-supervised learning to handle visually similar products. The model first detects and crops objects, then uses contrastive learning to extract better feature representations. This improves fine-grained classification, but adds computational complexity and depends on accurate detections.

Automated Checkout Systems (YOLOv4). James et al. [18] propose a checkout system using YOLOv4 for real-time product detection, counting, and billing. The system uses a pre-trained model and a simple pipeline to generate bills automatically. While it works well for known products, its performance drops for unseen or visually similar items and is slower on CPU.

Limitations of Existing Methods. Despite significant progress, existing approaches exhibit several important limitations. First, most methods rely heavily on supervised learning, which requires large labeled datasets and frequent retraining when new product categories are introduced. Second, these approaches struggle to generalize to unseen products, limiting their applicability in dynamic retail environments. Third, their performance degrades in real-world scenarios characterized by cluttered scenes, occlusions, and high visual similarity among products. Finally, the lack of scalability and adaptability makes these methods less suitable for practical deployment in continuously evolving retail systems.

TABLE I
COMPARISON OF PROPOSED APPROACH WITH BASELINE METHOD

Method	**ACD** (↓)	**mCCD** (↓)	**mCIoU** (↑)	**mAP@50** (↑)
Existing Methods				
Baseline Paper [10]	5.94	0.34	70.50	80.98
Proposed Methods				
YOLOv8	2.87	0.96	65.67	0.865
+ VAE	**0.13**	**0.22**	**66.97**	0.877
+ DINOv2	2.87	0.36	63.13	**0.921**

The proposed approach addresses the limitations of existing methods by shifting from traditional classification-based frameworks to an embedding-based similarity search paradigm. Unlike prior works that require retraining for new product categories, our method enables scalability by simply adding new embeddings to the database. Furthermore, the use of DINOv2 improves feature representation by capturing global contextual information, allowing better discrimination between visually similar products and improved robustness to occlusions. Experimental results demonstrate that our approach achieves superior performance in precision, recall, and mAP compared to baseline methods, making it more suitable for real-world retail checkout systems.

III. Proposed Methodology

Overall System Architecture. The proposed system follows a two-stage pipeline that combines object detection with embedding-based similarity search for retail product identification, as shown in Fig. 1. Given an input image containing multiple products, the system first detects individual objects and extracts them as cropped images. These crops are then converted into vector embeddings, which are stored in a database during training. During inference, embeddings of query products are matched with stored embeddings using similarity search to identify the most similar product.

In this work, we evaluate three approaches: (1) YOLOv8 as a baseline model for direct detection and classification, (2) YOLO combined with VAE-based embeddings, and (3) YOLO combined with DINOv2 embeddings for similarity-based retrieval.

Object Localization via YOLOv8. The framework begins with object detection using YOLOv8, a single-stage detector

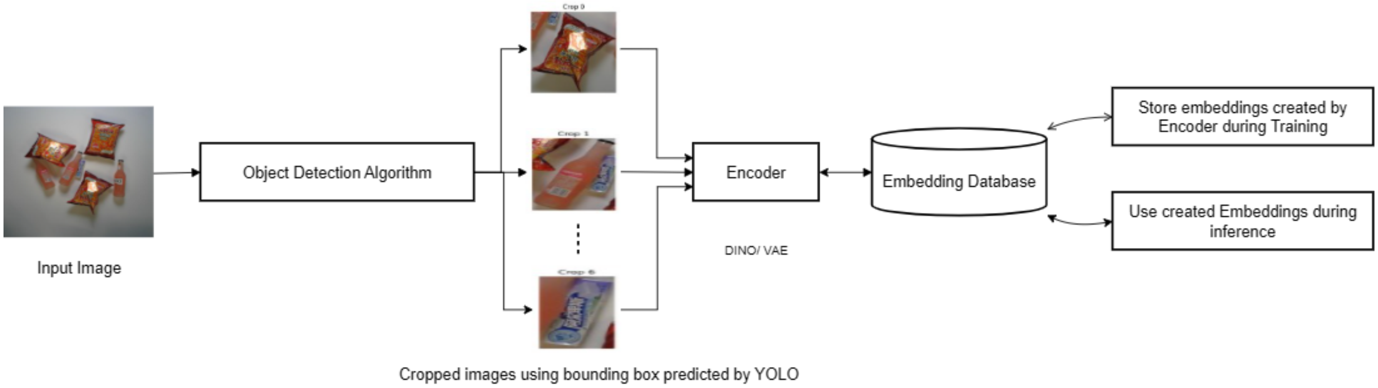

Fig. 1. Proposed Architecture

that predicts bounding boxes and confidence scores in one pass. For an input image I, the model produces detections of the form:

$$B = (x_c, y_c, w, h, s) \tag{1}$$

where (x_c, y_c) represent the center coordinates, (w, h) denote width and height, and s is the confidence score. Non-Maximum Suppression (NMS) is applied to remove redundant detections. The resulting bounding boxes are used to crop individual product images for further processing.

Feature Extraction via Variational Autoencoder (VAE). To generate compact representations of products, we use the encoder part of a Variational Autoencoder (VAE). The encoder maps each input crop x to a latent distribution defined by mean μ and variance σ^2:

$$q_\phi(z|x) = \mathcal{N}(z; \mu, \sigma^2) \tag{2}$$

A latent vector z is sampled using the reparameterization trick:

$$z = \mu + \sigma \odot \epsilon, \quad \epsilon \sim \mathcal{N}(0, I) \tag{3}$$

The VAE is trained using the loss:

$$\mathcal{L} = \|x - \hat{x}\|^2 + D_{KL}\big(q(z|x) \,\|\, p(z)\big) \tag{4}$$

The encoder learns latent embeddings that capture overall visual structure. However, since the objective focuses on reconstruction, the embeddings may not capture fine-grained differences between visually similar products effectively.

YOLO + DINOv2 (Vision Transformer-Based Encoder). To overcome the limitations of VAE, we use DINOv2 as an alternative encoder. DINOv2 is a self-supervised Vision Transformer (ViT)-based model that learns representations by modeling relationships between different parts of the image using self-attention.

The attention mechanism is defined as:

$$\text{Attention}(Q, K, V) = \text{Softmax}\left(\frac{QK^T}{\sqrt{d}}\right) V \tag{5}$$

DINO uses a teacher-student training framework where the student network learns to match the output distribution of a teacher network. This helps the model learn more discriminative and robust visual features without requiring labels. In our experiments, DINO-based embeddings were better at distinguishing similar-looking products compared to VAE.

Similarity Search using k-Nearest Neighbors (k-NN). Once embeddings are generated, product identification is performed using similarity search. Given a query embedding z_q and stored embeddings $\{z_i\}$, similarity is computed using cosine similarity:

$$\text{sim}(z_q, z_i) = \frac{z_q \cdot z_i}{\|z_q\| \|z_i\|} \tag{6}$$

We use k-Nearest Neighbors (k-NN) with $k = 5$ to retrieve the most similar embeddings. The final prediction is obtained using majority voting:

$$\hat{y} = \text{mode}(y_1, y_2, \ldots, y_k) \tag{7}$$

End-to-End Pipeline. The complete system operates as follows: an input image is processed by YOLOv8 to detect and crop individual products. Each crop is then passed through an encoder (VAE or DINOv2) to generate embeddings. During training, embeddings are stored in a database along with their labels. During inference, embeddings of detected products are compared with stored embeddings using k-NN similarity search, and the final label is assigned through majority voting.

This formulation converts the problem from direct classification to similarity-based retrieval, making the system more flexible and allowing new products to be added without retraining the model.

IV. Experimental Setup

Computational Environment. All experiments were conducted using the Kaggle notebook environment with GPU T4 acceleration. The implementation was done in Python using PyTorch for model development and the Ultralytics YOLOv8 framework for object detection.

Model Training and Implementation. In this work, we evaluate three different approaches for the retail checkout problem:

- YOLOv8 as a baseline model for direct detection and classification
- YOLOv8 + VAE for embedding-based retrieval
- YOLOv8 + DINOv2 for improved embedding-based retrieval

The YOLOv8 model was fine-tuned on the RPC dataset for object detection. For the second approach, a Variational Autoencoder (VAE) was trained using cropped product images to learn latent representations. For the third approach, pretrained DINOv2 models were used to extract high-quality feature embeddings without requiring labeled data.

Integration of Models. All components were integrated into a unified pipeline. YOLOv8 is used to detect and crop product regions. These cropped images are then passed through either the VAE encoder or the DINOv2 model to generate embeddings. The embeddings are stored in a database and queried during inference using FAISS for efficient similarity search.

Inference and Evaluation. During inference, YOLOv8 first detects product regions in the input image. Each detected crop is converted into an embedding using either VAE or DINOv2. The embedding is then compared with stored embeddings using cosine similarity, and classification is performed using k-Nearest Neighbors ($k = 5$) with majority voting.

Evaluation Metrics. The performance of all three approaches was evaluated using metrics defined in the RPC benchmark, including Absolute Count Difference (ACD), mean Category Count Difference (mCCD), and mean Category Intersection over Union (mCIoU). These metrics allow evaluation of both classification accuracy and counting performance in multi-object scenes.

A. *Dataset*

The experiments were conducted on the **Retail Product Checkout (RPC) dataset**, which is designed to simulate real-world checkout scenarios. The dataset contains both single-product images (used for training embedding models) and multi-product checkout images (used for validation and testing).

The dataset includes 200 product categories with variations in viewpoint, lighting, and occlusion. A key challenge in this dataset is the domain shift between training and testing data, where models trained on isolated product images must generalize to cluttered checkout scenes.

Additionally, the dataset provides annotations in JSON format and categorizes checkout images into three difficulty levels: *easy*, *medium*, and *hard*, enabling detailed evaluation under different levels of complexity.

The dataset is publicly available at: https://www.kaggle.com/datasets/diyer22/retail-product-checkout-dataset.

V. Results

A. *YOLOv8 Performance.*

The initial performance of the YOLOv8 model is documented in Fig. 2 and Fig. 3. Visual inspection of Fig. 2 confirms that YOLOv8 successfully localizes a variety of products within cluttered scenes, maintaining reliability across different item orientations and lighting shifts. The model exhibits precise localization, as seen in the tight bounding boxes around detected items. However, challenges arise in densely packed areas where objects overlap; we observed occasional duplicate detections and misclassifications, particularly among items with similar packaging designs. The quantitative breakdown in the Precision–Recall curve (Fig. 3) shows an overall mAP@0.5 of approximately 0.866. While the model maintains high precision for high-confidence predictions, precision naturally dips as recall increases and the system encounters more ambiguous or partially hidden instances. While YOLOv8 serves as a robust baseline for detection, its struggle with fine-grained visual details justifies the move toward the embedding-based system proposed here.

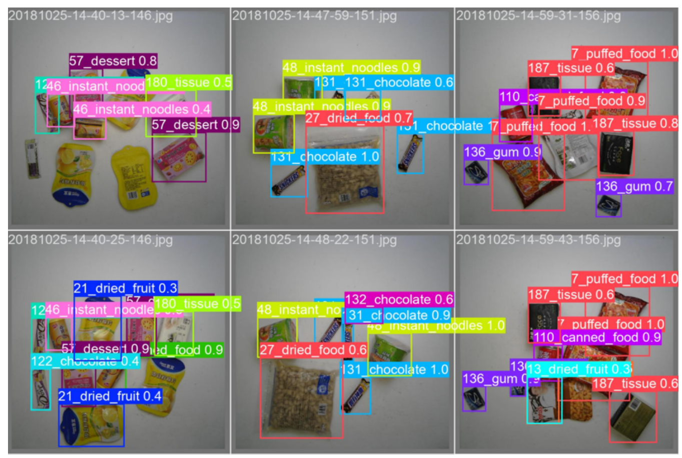

Fig. 2. YOLOv8 detection results showing multiple products in cluttered retail scenes with bounding boxes and class labels.

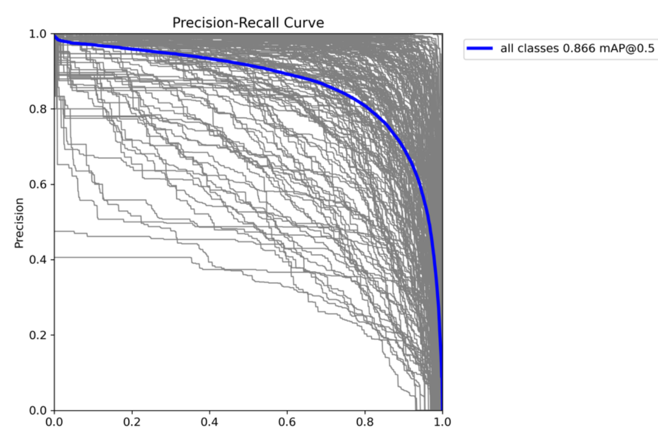

Fig. 3. Precision versus Recall curve for YOLOv8 showing overall detection performance with mAP@0.5 of 0.866.

B. *YOLOv8 + VAE Results.*

The performance of the YOLOv8 + VAE approach is illustrated in Fig. 5 and Fig. 6. As shown in Fig. 5, the VAE

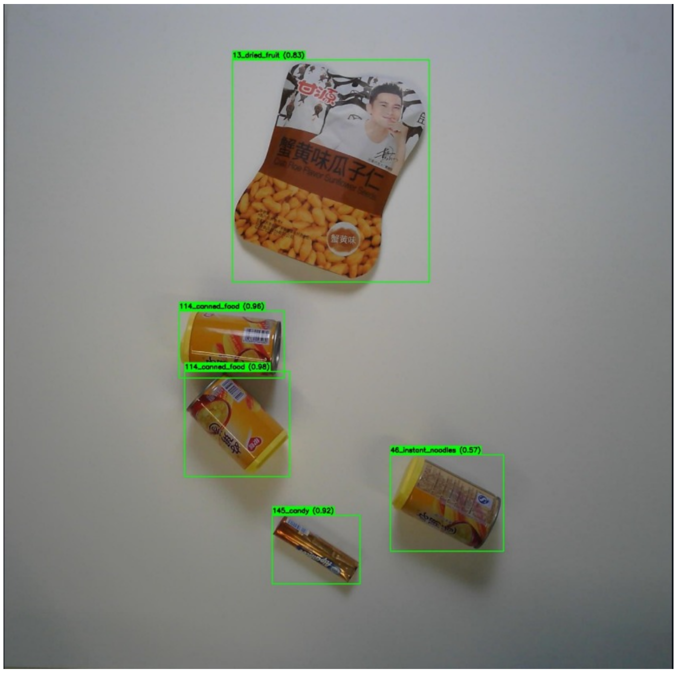

Fig. 4. Predictions Given by YOLO

training loss steadily decreases over epochs, indicating stable convergence and effective learning of latent representations. The loss curve shows a rapid decline in the initial epochs followed by gradual stabilization, suggesting that the model successfully captures the underlying distribution of product images. The qualitative detection and identification performance is further reflected through embedding-based matching, where the VAE encoder generates compact latent vectors for similarity comparison. The quantitative evaluation using the Precision–Recall curve in Fig. 6 demonstrates that the model maintains relatively high precision across moderate recall values, indicating reliable detection performance. However, a noticeable drop in precision is observed at higher recall levels, which suggests challenges in handling difficult cases and visually similar products. Compared to the baseline YOLOv8, the VAE-based approach provides improved feature representation; however, its reconstruction-driven objective limits its ability to capture highly discriminative features, resulting in lower performance compared to the DINOv2-based approach..

C. YOLOv8 + DINOv2 Performance.

The outcomes for the YOLOv8 + DINOv2 framework are shown in Fig. 7. Integrating DINOv2 yields a noticeable improvement in the system's ability to differentiate between similar products in messy environments. While the bounding box accuracy remains consistent with the YOLOv8 baseline, the embedding-based retrieval from DINOv2 provides far more reliable classification, even when items are partially blocked. This version of the model shows fewer errors in densely packed displays compared to previous iterations. The Precision–Recall curve in Fig. 8 supports these observations, showing a higher mAP@0.5 of roughly 0.94. Notably, the model sustains high precision across a broader recall range, proving it can handle difficult samples without sacrificing accuracy. We attribute this success to the DINOv2 Vision Transformer architecture, which uses self-attention to analyze the global context of the image. This makes the YOLOv8 + DINOv2 combination the most effective choice for actual retail deployment.

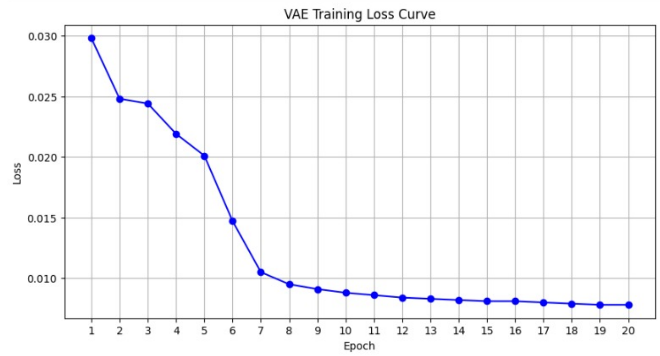

Fig. 5. Training loss curve of the Variational Autoencoder showing convergence over epochs.

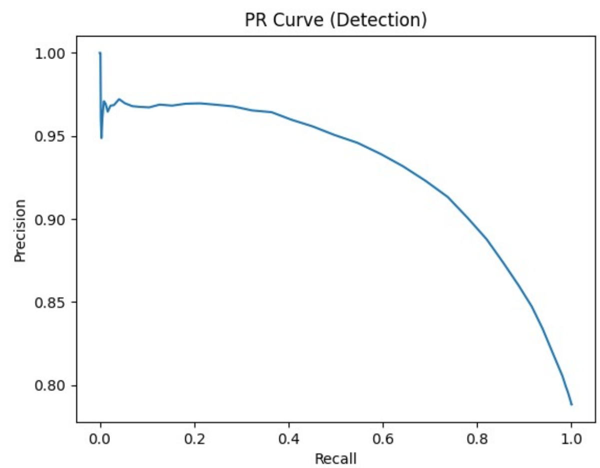

Fig. 6. Precision versus Recall curve for YOLOv8 + VAE showing detection performance.

Table II provides a direct numerical comparison. While the baseline YOLOv8 reaches a precision of 0.777, the DINOv2-enhanced version achieves the highest marks with 0.815 precision and 0.921 mAP@0.5. We also tracked inference speed using Frames Per Second (FPS). The standalone YOLOv8 model is clearly the fastest at 28.57 FPS, which is ideal for high-speed real-time use. Adding the embedding layers naturally creates a computational bottleneck; the VAE version runs at 4.65 FPS, while the more complex DINOv2 version drops to 3.12 FPS. This represents a clear trade-off: DINOv2 offers superior accuracy and identification at the cost of processing speed. However, for most retail checkout applications, the gain in accuracy is generally worth the slower frame rate. Additional localization metrics are also presented

Fig. 7. YOLOv8 + DINOv2 results showing improved product identification in cluttered retail scenes.

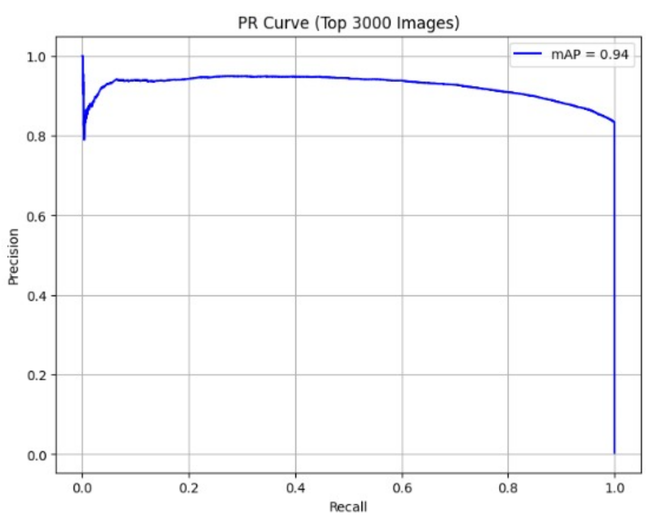

Fig. 8. Precision versus Recall curve for YOLOv8 + DINOv2 showing improved performance with mAP@0.5 of 0.94.

including Average Center Distance (ACD), mean Category Center Distance (mCCD), and mean Complete Intersection over Union (mCIoU). Lower ACD and mCCD values signify better alignment between the predicted and actual centers of the products. Meanwhile, higher mCIoU scores reflect better overall bounding box quality in terms of shape and overlap. While YOLOv8 + DINOv2 excels in classification metrics (precision, recall, mAP), the VAE-based approach shows competitive performance in localization-related metrics such as mCIoU.

TABLE II
PERFORMANCE COMPARISON

Metric	YOLOv8	YOLOv8 + DINOv2	YOLOv8 + VAE
Precision	0.777	**0.815**	0.788
Recall	0.784	**0.818**	0.793
mAP@0.5	0.865	**0.921**	0.877
mAP@0.5:0.95	0.714	0.634	**0.737**
FPS	**28.57**	3.12	4.65
ACD ↓	2.87	**2.87**	0.13
mCCD ↓	0.96	0.36	**0.22**
mCIoU ↑	**65.67**	63.13	66.97

VI. Discussion and Future Work

While the YOLOv8 + DINOv2 framework achieves superior performance in terms of accuracy and robustness, it introduces increased computational complexity and reduced inference speed compared to the baseline YOLOv8 model. This highlights a fundamental trade-off between accuracy and efficiency, which is critical for real-time retail deployment scenarios.

A key limitation of the current pipeline is its strong dependency on the object detection stage. Since the system relies on YOLOv8 to generate object crops, any missed detection directly leads to missed product identification. This creates a single point of failure, where undetected objects are never processed by the embedding-based retrieval module. Addressing this limitation is an important direction for future work.

One possible solution is to incorporate a complementary global retrieval mechanism that operates directly on the full image or dense region proposals, ensuring that missed detections can still be recovered through similarity matching. Alternatively, multi-stage detection strategies, ensemble detectors, or region proposal networks can be integrated to improve recall and reduce missed objects.

Future work can also focus on improving efficiency and scalability. Approximate Nearest Neighbor (ANN) methods such as FAISS can be further optimized to handle large-scale embedding databases with low latency. Additionally, lightweight transformer architectures or model compression techniques can be explored to accelerate inference without significantly sacrificing accuracy.

Another promising direction is the integration of multi-modal information. Combining visual embeddings with textual metadata (e.g., product names, barcodes, or descriptions) can enhance discrimination between visually similar products.

Finally, extending the system toward open-set recognition remains an important challenge. Enabling the model to identify and adapt to previously unseen products would significantly improve its applicability in dynamic retail environments. Improving robustness under extreme occlusions and domain shifts is also a key area for further research.

VII. Conclusion

In this work, we proposed a scalable and efficient retail product checkout system by integrating object detection with embedding-based similarity search. The system utilizes

YOLOv8 for accurate object detection and combines it with two feature extraction approaches, Variational Autoencoder (VAE) and DINOv2, to generate meaningful visual embeddings for product identification.

The experimental results demonstrate that while the baseline YOLOv8 model provides strong detection performance, incorporating embedding-based methods significantly enhances the system's ability to distinguish visually similar products. The VAE-based approach improves representation learning to some extent; however, its reconstruction-based objective limits its discriminative capability. In contrast, the DINOv2-based approach, leveraging Vision Transformer and self-attention mechanisms, achieves superior performance by capturing global contextual information and fine-grained visual features. This is reflected in higher precision, recall, and mAP scores.

A key advantage of the proposed system is its flexibility and scalability. Unlike traditional classification-based approaches, the embedding-based framework allows new products to be added without retraining the entire model, making it highly suitable for real-world retail environments where product inventories frequently change.

Overall, the proposed YOLOv8 + DINOv2 approach provides an effective solution for automatic retail checkout by improving accuracy, handling visually similar products, and enabling efficient similarity-based retrieval. Future work may focus on improving inference speed, optimizing embedding search using advanced indexing techniques, and extending the system to real-time deployment scenarios.

VIII. Evaluation Metrics

In this work, we evaluate the performance using both RPC-specific metrics for the automatic checkout (ACO) task and standard object detection metrics.

Let N denote the number of checkout images and K denote the number of product categories (SKUs). Let $P_{i,k}$ and $GT_{i,k}$ represent the predicted and ground truth counts of category k in image i, respectively.

A. RPC Metrics

a) Average Counting Distance (ACD).: ACD computes the average counting error per image:

$$ACD = \frac{1}{N} \sum_{i=1}^{N} \sum_{k=1}^{K} CD_{i,k} \tag{8}$$

b) Mean Category Counting Distance (mCCD).: mCCD measures the normalized counting error across categories:

$$mCCD = \frac{1}{K} \sum_{k=1}^{K} \frac{\sum_{i=1}^{N} CD_{i,k}}{\sum_{i=1}^{N} GT_{i,k}} \tag{9}$$

c) Mean Category Intersection over Union (mCIoU).: mCIoU evaluates the overlap between predicted and ground truth counts:

$$mCIoU = \frac{1}{K} \sum_{k=1}^{K} \frac{\sum_{i=1}^{N} \min(GT_{i,k}, P_{i,k})}{\sum_{i=1}^{N} \max(GT_{i,k}, P_{i,k})} \tag{10}$$

B. Object Detection Metrics

In addition to RPC metrics, we report standard object detection metrics.

a) Precision.:

$$Precision = \frac{TP}{TP + FP} \tag{11}$$

b) Recall.:

$$Recall = \frac{TP}{TP + FN} \tag{12}$$

c) Mean Average Precision at IoU 0.5 (mAP@0.5).:

$$mAP@0.5 = \frac{1}{C} \sum_{c=1}^{C} AP_c \tag{13}$$

where C is the number of classes.

d) Mean Average Precision (mAP@0.5:0.95).:

$$mAP@0.5:0.95 = \frac{1}{10} \sum_{t=0.5}^{0.95} mAP@t \tag{14}$$

where t varies from 0.5 to 0.95 with a step size of 0.05.

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